Solution

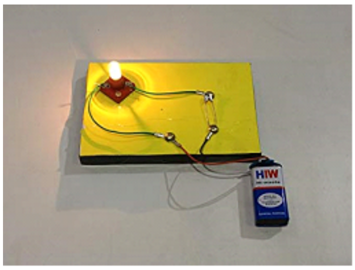
ELECTRICITY -03

Class 10 - Science

Section A

1. **Read the text carefully and answer the questions:**

How do we express electric current? Electric current is defined by the amount of charge flowing through a particular area in unit time. In other words, it is the rate of flow of electric charges. In circuits using metallic wires, electrons constitute the flow of charges. However, electrons were not known at the time when the phenomenon of electricity was first observed. So, electric current was considered to be the flow of positive charges and the direction of flow of positive charges was taken to be the direction of electric current. Conventionally, in an electric circuit, the direction of electric current is taken as opposite to the direction of the flow of electrons, which are negative charges.



(i) If a net charge Q, flows across any cross-section of a conductor in time 't', then the current 'I', through the cross-section is given by I = Q/T

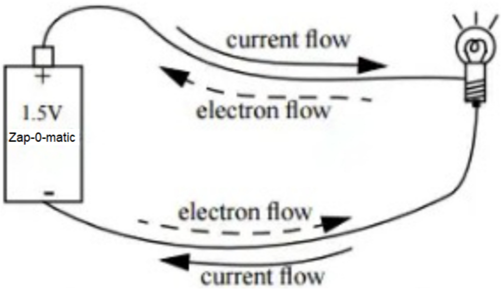
(ii) The SI unit of electric charge is the coulomb (C), which is equivalent to the charge contained in nearly 6×10^{18} electrons.

(iii) The electric current is expressed by a unit called ampere (A). One ampere is constituted by the flow of one coulomb of charge per second.

(iv) $I = \frac{Q}{t} = \frac{12}{4} = 3A$

2. **Read the text carefully and answer the questions:**

The rate of flow of charge is called electric current. The SI unit of electric current is Ampere (A). The direction of flow of current is always opposite to the direction of flow of electrons in the current.



The electric potential is defined as the amount of work done in bringing a unit-positive test charge from infinity to a point in the electric field. The amount of work done in bringing a unit positive test charge from one point to another point in an electric field is defined as potential difference.

$V_{AB} = V_B - V_A = \frac{W_{BA}}{q}$

The SI unit of potential and potential difference is volt.

(i) $V = \frac{W}{q} = \frac{W}{It}$

(ii) I = 1 A, t = 1 s

q = It = 1 $\times$ 1 = 1C

$n = \frac{q}{e} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$

(iii) The potential difference is the work done in moving a unit of positive electric charge from one point to another.

W = 100 J, q = 20 C

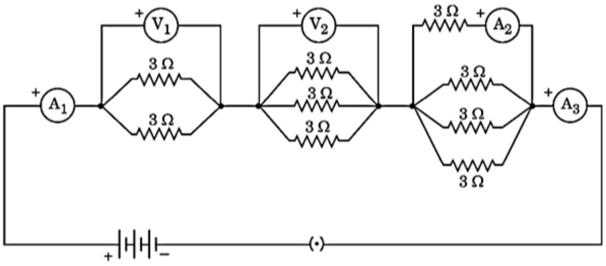
$V = \frac{W}{q} = \frac{100}{20} = 5$ V

(iv) q = 2 C, t = 100 ms = 0.1 s

$I = \frac{q}{t} = \frac{2}{0.1}$ = 20 A

3. **Read the text carefully and answer the questions:**

Consider the following electrical circuit diagram in which nine identical resistors of 3 Ω each are connected as shown. If the reading of the ammeter A_1 is 1 ampere, answer the following questions:



(i) i. Both have same reading i.e $A_1 = A_3$

ii. Both are connected in series

(ii) Reading of $A_2 = \frac{1}{4}$ A_3 as current is equally divided in the four identical resitors i.e ($A_2 < A_3$).

(iii) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ Or $R_P = \frac{R}{n}$

$\frac{1}{R_P} = \frac{1}{3\Omega} + \frac{1}{3\Omega}$,$R_P = \frac{3}{2}\Omega$

V = I R

$V_1 = 1A \times \frac{3}{2}\Omega = \frac{3}{2}V = 1.5V$

(iv) $\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega}$

$\therefore R_p = \frac{3}{2}\Omega$

$\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega} + \frac{1}{3\Omega}$

$\therefore R_{p_2} = 1\Omega$

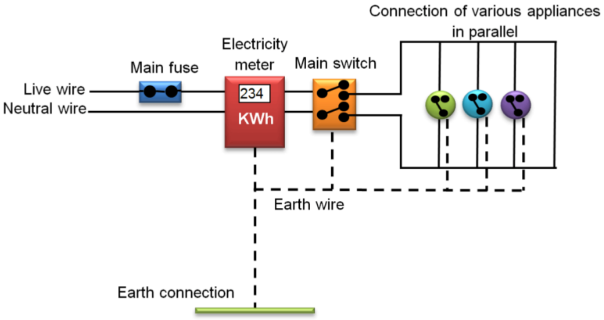
$\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega} + \frac{1}{3\Omega} + \frac{1}{3\Omega}$

$\therefore R_{p_3} = \frac{3}{4}\Omega$

$\therefore R = R_{p_1} + R_{p_2} + R_{p_3} = \left(\frac{3}{2} + 1 + \frac{3}{4}\right)\Omega = \frac{13}{4}\Omega = 3.25\Omega$

4. **Read the text carefully and answer the questions:**

In our homes, either the overhead electric poles or underground cables support the power supply flowing through the mains supply. One of the wires in this supply is covered with insulation in the colour red, and another wire colored black. At the meter board, these wires pass into an electric meter through the main fuse. The main switch, live wire, and the neutral wire are in connection to the line wires in our homes; these wires then supply electricity to separate electric circuits within the house.



(i) Live wire is of Red colour.

(ii) The fuse is connected in between live wire.

(iii) KWh is the commercial unit of power supply.

(iv) A fuse wire is a safety device connected in series with the live wire of circuit. It has high resistivity and a low melting point.

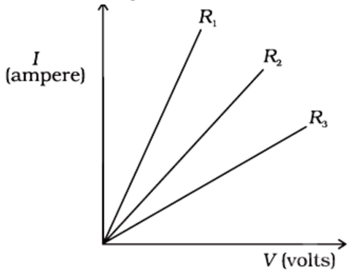
5. **Read the text carefully and answer the questions:**

In 1827, a German physicist Georg Simon Ohm (1787-1854) found out the relationship between the current I, flowing in metallic wire and the potential difference across its terminals. He stated that the electric current flowing through a metallic wire is directly proportional to the potential difference V, across its ends provided its temperature remains the same.

The resistance of a circuit is defined as the ratio between the voltage applied to the current flowing through it. Rearranging the above relation,

$R = \frac{V}{I}$

Electric charge flows easily through some materials than others. The electrical resistance measures how much the flow of this electric charge is restricted within the circuit.



(i) Ohm is the unit of electrical resistance.

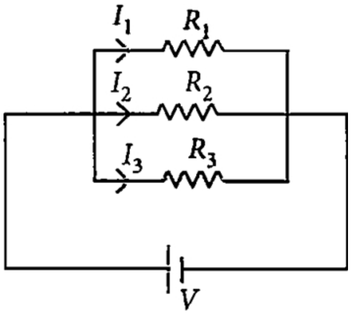
(ii) According to Ohm's law, there is a relation between the current flowing through a conductor and the potential difference across it. It is given by,
V ∝ I V = IR

(iii) R_3 resistance has high resistance.

(iv) The slope of V-I graph at any point represents resistance.

6. **Read the text carefully and answer the questions:**

If two or more resistances are connected in such a way that the same potential difference gets applied to each of them, then they are said to be connected in parallel.



The current flowing through the two resistances in parallel is, however, not the same. When we have two or more resistances joined in parallel to one another, then the same current gets additional paths to flow and the overall resistance decreases.

(i) The equivalent resistance in the parallel combination is lesser than the least value of the individual resistance.
The equivalent resistance of parallel combinations
$\frac{1}{Rp} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$
$\Rightarrow$ Rp = $\frac{8}{7}\Omega$
Thus equivalent resistance is less than 2Ω.

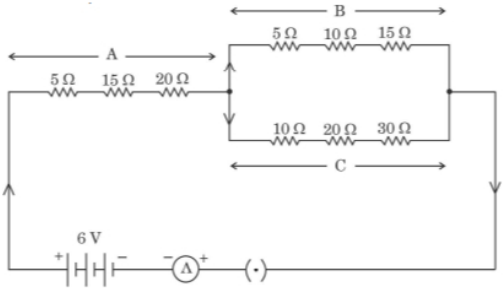
(ii) Resistance of each piece $= \frac{12}{3} = 4\Omega$
$\frac{1}{R_p} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \Rightarrow R_p = \frac{4}{3}\Omega$

(iii) All the three resistors are in paralle.
$\therefore \frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} + \frac{1}{1} = \frac{1+2+6}{6} = \frac{9}{6}\ R_P = \frac{6}{9} = \frac{2}{3}\Omega$

(iv) All are in parallel.
$\frac{1}{R_p} = \frac{1}{12} \times 4 = \frac{1}{3} \Rightarrow R_p = 3\Omega$
$I = \frac{3}{3} = 1\ \text{A}$
So, current in each resistor $I' = \frac{3}{12} = \frac{1}{4}\ \text{A}$

7. **Read the text carefully and answer the questions:**

Study the following electric circuit in which the resistors are arranged in three arms A , B and C:



(i) Since the resistance in arm B are connected in series.
So, R_B = $5\Omega + 10\Omega + 15\Omega$

$R_B = 30\Omega$

(ii) Total resistance in arm C

$R_C = 10\Omega + 20\Omega + 30\Omega$

$R_C = 60\Omega$

Now as arm B and arm C are in parallel

Equivalent resistance $\frac{1}{R} = \frac{1}{R_B} + \frac{1}{R_C}$

$\frac{1}{R} = \frac{2+1}{60} = \frac{3}{60}$

$R = 20\Omega$

(iii) Total resistance in arm A

$R_A = 5\Omega + 15\Omega + 20\Omega$

$R_A = 40\Omega$

Now, Equivalent resistance of circuit

$R_{eq} = R_{eq} + R$

$R_{eq} = 40 + 20 = 60\Omega$

By ohm's law

$V = IR$

$I = \frac{V}{R} = \frac{6}{60}$ = 0.1 A

(iv) If arm B is removed

Equivalent resistance in circuit

$R_{eq} = R_A + R_C$

$R_{eq} = 40 + 60 = 100\Omega$

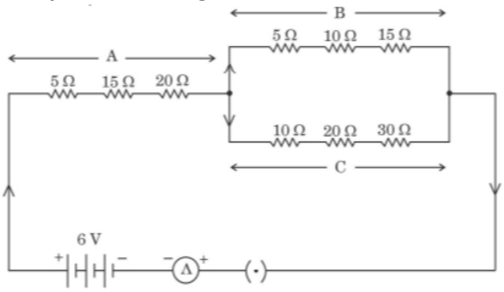
From ohm's law

$V = IR$

$I = \frac{V}{R} = \frac{6}{100}$ = 0.06 A

8. **Read the text carefully and answer the questions:**

Study the following electric circuit in which the resistors are arranged in three arms A, B and C:



(i) ∵ In arm A three resistance are connected in series

So,

$R_s = R_1 + R_2 + R_3$

$R_s = 5\Omega + 15\Omega + 20\Omega$

$R_s = 40\Omega$

(ii) For arm B

As resistance are connected in series

$R_B = R_1 + R_2 + R_3$

$R_B = 5 + 10 + 15 = 30\ \Omega$

For arm C

As resistance are in series

$R_C = R_1 + R_2 + R_3$

$R_C = 10 + 20 + 30$

$R_C = 60\ \Omega$

Now, equivalent resistance of arm B and arm C

$\frac{1}{R} = \frac{1}{R_B} + \frac{1}{R_C}$

$\frac{1}{R} = \frac{1}{30} + \frac{1}{60} = \frac{2+1}{60} = \frac{3}{60}$

R = 20 Ω

(iii) Total resistance of circuit

$R_{equi} = R_s + R$

$R_{equi} = 40 + 20 = 60\Omega$

∵ V = IR

I = $\frac{V}{R}$

I = $\frac{6}{60} = \frac{1}{10}$

I = 0.1 A

(iv) If arm B is removed

Total resistance = 40 + 60 = 100Ω

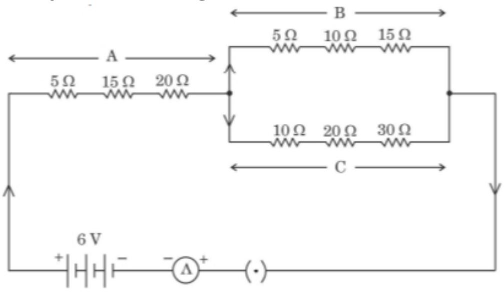
I = $\frac{V}{R}$

I = $\frac{6}{100}$

I = 0.06 A

9. **Read the text carefully and answer the questions:**

Study the following electric circuit in which the resistors are arranged in three arms A, B and C :



(i) ∵ Resistance in arms C are in series

So, Equivalent resistance of arm C

$R_c = 10 + 20 + 30$

$R_c = 60\Omega$

(ii) Since resistance in arm B are in series

So, total resistance in arm B

$R_B = 5 + 10 + 15 = 30\Omega$

Now, Equivalent resistance of arm B and C

$\frac{1}{R} = \frac{1}{R_B} + \frac{1}{R_c}$

$\frac{1}{R} = \frac{1}{30} + \frac{1}{60}$

$\frac{1}{R} = \frac{2+1}{60} = \frac{3}{60}$

R = 20Ω

(iii) Total resistance in arm A, as resistance are in series

$R_A = 5 + 15 + 20 = 40\Omega$

Equivalent resistance in circuit

$R_{eq} = R_A + R$

$R_{eq} = 40 + 20 = 60\ \Omega$

By ohm's law,

V = IR

$I = \frac{V}{R} = \frac{6}{60}$ = 0.01 A

(iv) If arm B is removed

Then total resistance of circuit

$R_{eq} = R_A + R_C$

$R_{eq} = 40 + 60 = 100\Omega$

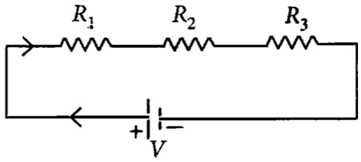
By ohm's law

V = IR

$I = \frac{V}{R} = \frac{6}{100}$ = 0.06 A

10. **Read the text carefully and answer the questions:**

Two or more resistances are connected in series or in parallel or both, depending upon whether we want to increase or decrease the circuit resistance.



The two or more resistances are said to be connected in series if the current flowing through each resistor is the same.

(i) In series combination, Rs = $R_1 + R_2 + R_3$ = R + R + R = 3R.

(ii) The equivalent resistance is where the total resistance is connected either in parallel or in series.

Resistance of each wire = $\frac{20}{4}$ = 5 Ω

Equivalent resistance in series

R_s = 5 + 5 + 5 + 5 = 20Ω

(iii) All are in series, R_s = 5R = 5 × 2 = 10Ω

(iv) R_s = 1 + 2 + 3 = 6 Ω

$I = \frac{18}{6} = 3$ A

11. **Read the text carefully and answer the questions:**

The heating effect of current is obtained by transformation of electrical energy into heat energy. Just as mechanical energy used to overcome friction is covered into heat, in the same way, electrical energy is converted into heat energy when an electric current flows through a resistance wire. The heat produced in a conductor, when a current flows through it is found to depend directly on (a) strength of current (b) resistance of the conductor (c) time for which the current flows.

The mathematical expression is given by H = I^2Rt.

The electrical fuse, electrical heater, electric iron, electric geyser etc. all are based on the heating effect of current.

(i) Low resistance, high melting point.

(ii) High resistance, low melting point

Electric Fuse is based on the principle of the heating effect of Electric current.

(iii) Given: H = I^2Rt

So, H' = $(2I)^2 \cdot \frac{R}{2}t$ = 2 H

(iv) Given: I = 5 A, resistance = R. Let r be the new radius.

Now, H = i^2Rt ...(a)

Also H' = I^2R' t ...(b)

From (a) and (b), $5^2 \times \rho\frac{L}{\pi r^2}t = 10^2 \times \rho\frac{L}{\pi r'^2} \cdot t$

$\frac{25}{r^2} = \frac{100}{r'^2} \Rightarrow \frac{r'}{r} = 2 \Rightarrow$ r' = 2r

12. **Read the text carefully and answer the questions:**

The electrical energy consumed by an electrical appliance is given by the product of its power rating and the time for which it is used. The SI unit of electrical energy is Joule. Actually, Joule represents a very small quantity of energy and therefore it is inconvenient to use where a large quantity of energy is involved. So for commercial purposes, we use a bigger unit of electrical energy which is called kilowatt-hour. 1 kilowatt-hour is equal to 3.6 × 10^6 joules of electrical energy.

(i) $E \propto t$

When the time of operating the heater is doubled, the energy dissipated is doubled.

(ii) Given: P = 60 W, t = 1 min

E = 60 × 1 × 60 = 3600 J

(iii) Given: P = 400 Ω, t = 8 hour

E = 400 × 8 = 3200 Wh = 3.2 kWh

Cost = 3.2 × 5 = ₹16

(iv) Given: I = 5A, R = 2Ω, t = 30 min

E = I^2Rt = 5 × 5 × 2 × 30 × 60

E = 90000 J = 90 kJ

13. **Read the text carefully and answer the questions:**

We know that a battery or a cell is a source of electrical energy. The chemical reaction within the cell generates the potential difference between its two terminals that sets the electrons in motion to flow the current through a resistor or a system of resistors connected to the battery. To maintain the current, the source has to keep expanding its energy. Where does this energy go? A part of the source energy in maintaining the current may be consumed for useful work (like in rotating the blades of an electric fan). The rest of the source energy may be expended in heat to raise the temperature of the gadget. We often observe this in our everyday life. For example, an electric fan becomes warm if used continuously for a long time, etc. On the other hand, if the electric circuit is purely resistive, that is, a configuration of resistors only connected to a battery; the source energy continually gets dissipated entirely in the form of heat. This is known as the heating effect of electric current. This effect is utilized in devices such as an electric heater, electric iron, etc.

(i) The law implies that heat produced in a resistor is
   a. directly proportional to the square of current for a given resistance,
   b. directly proportional to resistance for a given current, and
   c. directly proportional to the time for which the current flows through the resistor.

(ii) Firstly, we calculate the current flowing through it, using the relation I = $\frac{V}{R}$. Then we apply the formula H = I^2Rt to calculate the heating effect.

(iii) Heat produced, H = VIt

(iv) The resistivity of nichrome is more than that of copper so its resistance is also high. Therefore, a large amount of heat is produced in the nichrome wire for the same current as compared to that of copper wire.

Section B

14.

(c) Watt

Explanation: Watt is the unit that is used for expressing the electric power.

15.

(c) 250 kWh

Explanation: Electricity consumption is measured in Kilowatt-hour which is commonly called a unit. So, 250 unit is equal to 250 kWh. Also, 1 kWh = 3600000 J or 3.6 Mega Joules (MJ).

16. **(a)** 4 A

Explanation: Given,

P = 920W

V = 230V

I = ?

We know that,

P = VI

920 = 230I

I = $\frac{920}{230}$

I = 4 A

17.

(b) volt ampere

Explanation: The rate at which the heat is dissipated is called the power and is measured in units of watts (W). To find the power (P) using V and I, we use P=VI.

Here, the unit of Voltage (V) is volt and Current (I) is ampere.

Hence, the unit of electric power can also be expressed as a **volt-ampere.**

18. **(a)** $\frac{4V}{R}$

Explanation: We know that,

P = $\frac{V2}{R}$

R is constant

V is doubled

Therefore, P = $\frac{2(V2)}{R}$

P = $\frac{4V}{R}$

Therefore, when the resistance is doubled, power becomes four times the actual value.

19.

(d) 6W

Explanation: We know that power is calculated as P = VI. Substituting the values of V and I we get P = 6W.

20.

(c) 25 W

Explanation: Resistance of the electric bulb is $R = \frac{V^2}{P}$ or $R = \frac{220^2}{100}$

When it is operated at 110 V, the power consumed will be $P = \frac{V^2}{R}$ $or\ P = \frac{110^2 \times (100)}{(220^2)}$ $or\ P = \frac{100}{4}$ $or\ P = 25W$

21. **(a)** 55

Explanation: For heater,

P = 2kWh

T = 4 hours

Energy consumed = PT = 8kWh

For TV,

P = 200 watt = 0.2kW

T = 4 hours

Energy consumed = PT = 0.8kWh

For lamps,

P = 100 watt = 0.1kW

T = 4 hours

n = 3

Energy consumed = PTn = 1.2kWh

Total energy consumed = 8 + 0.8 + 1.2 = 10kWh

Cost of 1kWh = Rupees 5.50

Total cost = Rupees 55

22.

(d) 1 **:** 4

Explanation: Since the two conducting wires of the same material (same ρ) and of equal lengths (same l) and equal diameters (same A), they will have the same resistance R. $R = \rho\frac{l}{A}$

The effective resistance of the series combination will be $R_S = R + R$ or $R_S = 2R$. Current drawn by the series combination will be $I_S = \frac{V}{2R}$

The effective resistance of the parallel combination will be $R_P = \frac{R \times R}{R + R}$ or $R_P = \frac{R}{2}$. Current drawn by the parallel combination will be $I_P = \frac{2V}{R}$

Therefore, at a given voltage, the ratio of the resistances will be R_S **:** R_P = 4 **:** 1 and the ratio of the currents will be I_S **:** I_P = 1 **:** 4

The ratio of heat produced in the series and the parallel combination would be**:**

$\frac{H_S}{H_P} = \frac{{I_S}^2 \times R_S \times t}{{I_P}^2 \times R_P \times t}$ **or** $\frac{H_S}{H_P} = (\frac{I_S}{I_P})^2 \times (\frac{R_S}{R_P})$ **or** $\frac{H_S}{H_P} = \frac{1}{4}$

23.

(c) 2.99 kW

Explanation: Given,

I = 13 amp

V = 230 V

P = VI

P = 2.99 kW

24. **(a)** Brightness of bulb B will be more than that of A

Explanation: Since bulbs are connected in parallel so the resistance of combination would be less than the arithmetic sum of the resistance of all the bulbs. So, there will be no negative effect on the flow of current. As a result, bulbs would glow according to their wattage.

Hence, the brightness of bulb B will be more than that of A.

25.

(c) 850 W

Explanation: The electric iron that is used in our homes have 850 W power.

26. **(a)** Only (iii)

Explanation: Given: V = 220 V, P = 60 W

$P=\frac{V^2}{R}$ ...(i)

As bulbs are connected in series, then

P' = $\frac{V^2}{R_{\text{eq}}}=\frac{V^2}{R+R}=\frac{V^2}{2R}$

or P' = $\frac{1}{2}P$ (From (i))

or P' = $\frac{60}{2}=30$ W

As bulbs are connected in parallel, then

$\frac{1}{\mathrm{R}_{\text{eq}}}=\frac{1}{R}+\frac{1}{R}$ or $R_{\text{eq}}=\frac{R}{2}$

So power $P_{\text{eq}}=\frac{V^2}{R_{\text{eq}}}=\frac{V^2}{R}\times 2$

or p_{eq} = 2P = 2 $\times$ 60 = 120 W

27.

(c) 2.2 kWh

Explanation: Given,

I = 5 amp

V = 220V

T = 2 hours

P = ?

E = ?

P = VI

P = 1.1kW

Energy consumed E = PT

E = 2.2 kWh

28.

(c) 300 %

Explanation: Let the original value of current is I ampere.

The new value of current $I'=I+100\%I=2I$

Now, the new power $P'=I'^2R=4I\cdot R=4P$

So, the change in power $=P'-P=3P$

Therefore, the change in power in terms of percentage is 300%.

29.

(b) 10 A

Explanation: the rating of the fuse is10 A

30.

(c) IR^2

Explanation:

Electrical power (P) is the rate at which electric energy is lost (dissipated) or consumed in an electrical device due to resistance (R). P $=V\times I=I^2\times R=\frac{V^2}{R}$ where V is the potential difference across the resistor and I is the current passing through the resistor.

31.

(b) 3 J/s

Explanation: When 6 V and 0.5 A lamp is used, 3 J/s electrical energy is required.

32.

(b) 5 A

Explanation: 5 A

33.

(d) $\frac{I^2}{R}$

Explanation: Power (P) is the rate at which electric energy is lost (dissipated) or consumed in an electrical device due to resistance (R).

P = VI = I^2R ($\because$ V = I $\times$ R)

P = VI = $\frac{V^2}{R}$($\because$ I = $\frac{V}{R}$)

34.

(d) $\frac{l_1}{l_2} = \frac{1}{2}$

Explanation: Here, resistance of wire, R = 12Ω

Equivalent resistance of lengths l_1 and $l_2 = \frac{8}{3}\Omega$

$\therefore R_{\text{eq}} = \frac{R_1R_2}{R_1+R_2}$

We know, R_1 + R_2 = R = 12Ω

or R_1R_2 = 32

From (i) and (ii)

or R ∞ l

So, $\frac{R_1}{R_2} = \frac{l_1}{l_2}$ or $\frac{l_1}{l_2} = \frac{1}{2}$

35.

(d) 1 : 1 : 1

Explanation: We know, current flowing through the wire,

$I = \frac{dq}{dt} = \frac{Q}{t}$

or Q = l $\times$ t

Thus, area under the graph between current and time will give charge flowing through the wire.

For first time interval, (t_1 = 2 - 1 = 1 s)

Q_1 = $l_1 \times t_1$ = 2 $\times$ 1 = 2 C

For second time interval, (t_2 = 5 - 3 = 2 s)

Q_2 = $l_2 \times t_1$ = 1 $\times$ 2 = 2 C

For thrid time interval, (t_3 = 8 - 6 = 2 s)

Q_3 = $\frac{1}{2} \times l_3 \times t_3$ = $\frac{1}{2} \times 2 \times 2$ = 2 C

$\therefore$ Ratio of charge flowing through the wire at different time intervals is

$Q_1 : Q_2 : Q_3$ = 2 : 2 : 2 = 1 : 1 : 1

36.

(b) Independent of x

Explanation: We know that, $R = \rho\frac{I}{A}$

Here, resistivity = ρ, l = x, A = xy

$\therefore$ R = $R = \rho\frac{1}{A} = \rho\frac{x}{xy}$

or R = $\frac{\rho}{y}$

Thus, resistance between two opposite faces, shown by the shaded areas in the figure is independent of x.